

Education

- ✧ **PhD in Industrial and Systems Engineering**, Virginia Tech *2023-Present*
- ✧ **MEng in Industrial and Systems Engineering**, Virginia Tech *2023-2024*
- ✧ **B.Sc. in Industrial Engineering**, Sharif University of Technology *2016-2021*

Selected Graduate Coursework

- ✧ Stochastic and Robust Optimization, Integer Programming, Linear Optimization, Non-linear Optimization, Random Processes, Simulation, Machine Learning, Management of Organizational Systems, Socio-technical Systems

Research Interests

- ✧ **Methodology:** Stochastic Optimization, CVaR Optimization, Machine Learning
- ✧ **Application:** Epidemic Management, Invasive Species Control, Healthcare Policy, Behavioral Economics

Publications

- ✧ Escobar, L., Jozani, K., Romero-Alvarez, D., Brennan, R.N., Ulloa-Zepeda, L., Vial, C., Buyuktahtakin, E. Epidemiology of Andes virus (Orthohantavirus andesense). Under Review at The Lancet.
- ✧ Jozani, K., Sageer, N. A., Eldardiry, H., Tunc, S., & Buyuktahtakin, E. (2025). A Multi-echelon Demand-driven Supply Chain Model for Proactive Optimal Control of Epidemics: Insights from a COVID-19 Study. IISE Transactions. (Preprint available on ArXiv).
- ✧ Choi, S. J., Jozani, K., Cooper, J. F., & Buyuktahtakin, E. (2025). Learning to Optimize at Scale: A Benders Decomposition-TransfORMers Framework for Stochastic Combinatorial Optimization. NeurIPS 2025 Workshop on Machine Learning and Operations Research (MLxOR).
- ✧ Galande, N., Jozani, K., & Buyuktahtakin, E. (2025). Artificial Intelligence in Supply Chain Optimization: A Systematic Review of Machine Learning Models, Methods, and Applications. Under review at Annals of Operations Research.
- ✧ Yan, K., Jozani, K., & Buyuktahtakin, E. (2025). AI-Driven Spatio-Temporal Forecasting and Management of Wavyleaf Grass. Workshop paper Presented at the 20th INFORMS Workshop on Data Mining and Decision Analytics (DMDA).

Ongoing Research

- ✧ Hantavirus Modeling, Joint work with Escobar, L., and Buyuktahtakin, E. (May 2026 - Present)
- ✧ Psychiatric setting operations simulation and optimal control, Joint work with Carilion Clinic (February 2026 - Present)
- ✧ Learning to decompose optimization problems. Joint work with Fioretto, F., and Buyuktahtakin, E. (August 2025 - Present).
- ✧ Sample complexity in CVaR problems. Joint work with Khodadadian, S., and Buyuktahtakin, E. (April 2025 - Present).
- ✧ Epidemic management under distributional uncertainty. Joint work with Tunc, S., and Buyuktahtakin, E. (March 2025 - Present).

Peer Review

- ✧ Invited Review for IISE Transactions journal *July 2026*
- ✧ Invited Review, Naval Research Logistics journal *April 2026*

Teaching Experience

- ✧ Instructor of record for Probability Foundations for ISEs, Virginia Tech, US *2025*
- ✧ Graduate Teaching Assistant, Virginia Tech, US *2023 – 2025*
- ✧ Mathematics Teacher, Pajooresh High School, Iran *2016- 2022*

Research Experience

- ✧ Systems Optimization and Machine Learning Lab (SysOptiMaL), Virginia Tech *Dec 2023 - Present*
 - Epidemic management: Large-scale, multi-echelon supply chain optimization *Dec 2023 - Aug 2025*
 - Epidemic management: Prediction and control of Avian Influenza outbreaks in US farms *Feb 2026 - Present*
 - AI-driven optimization: Developing a learning-to-decompose framework *Aug 2025 - Present*
 - AI-driven optimization: Learning-enhanced optimization algorithm design *May 2024 - Present*
 - Invasive species management: Optimal control of Wavyleaf in Virginia *Aug 2025 - Present*
 - Healthcare management: Assessment of no-shows and cancellations in psychiatric settings *Feb 2026 - Present*
- ✧ Human Technology Interaction Research, Virginia Tech *Mar 2022 - Dec 2023*
- ✧ Consumer Behavior Analysis During COVID-19, Sharif University of Technology *Jun 2020 - Feb 2021*

Academic Presentations

- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Decision-Dependent Distributionally Robust Optimization Model for Proactive Mitigation of Infectious Disease Outbreaks.” INFORMS 2026 Annual Meeting Job Market Showcase Track. November 2026.
- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Decision-Dependent Distributionally Robust Optimization Model for Proactive Mitigation of Infectious Disease Outbreaks.” MSOM Conference. July 2026.
- ✧ Jozani, K., Khodadadian, S., Buyuktahtakin, E. “A General Framework for Finite-Sample Evaluation of CVaR Optimization.” INFORMS 2026 Optimization Society Conference. March 2026.
- ✧ Jozani, K., Choi, S. J., Cooper, J. F. Buyuktahtakin, E. “Learning to Optimize at Scale: A Decomposition-TransfORMers Framework for Stochastic Combinatorial Optimization.” 2025 NeurIPS MLxOR Workshop. December 2025.
- ✧ Jozani, K., Tunc, S., Buyuktahtakin, E. “A Multi-Stage Dynamic Optimization Framework for Effective Epidemic Resource Allocation and Sustained Public Trust.” INFORMS 2025 Annual Meeting. October 2025.
- ✧ Jozani, K. “Proactive and Need-based Epidemic Resource Allocation Through an Integrated Supply Chain Framework: Insights from a COVID-19 Study.” Invited Presentation at the David Eccles School of Business at The University of Utah. September 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” INFORMS MIF 2025 Summer Webinar Competition. July 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” VT GPSS Research Symposium. March 2025.

- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” VT Montgomery Distinguished Lecture Poster Presentation. March 2025.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” INFORMS 2024 Annual Meeting Health Applications Society Invited Presentation. October 2024.
- ✧ Jozani, K., Buyuktahtakin, E. “Equitable Vaccine Allocation through a Dynamic Epidemic-Supply Chain Model.” 2024 CeZAP Infectious Diseases Research Symposium. October 2024.

Grants

- ✧ Track 2 Seed Grant from Whole Health Consortium at Virginia Tech (Student PI) *2026*
- ✧ Invasive Species Collaborative (ISC) at Virginia Tech Graduate Student Grant Program (co-PI) *2025-2026*
- ✧ VT Center for Emerging, Zoonotic, and Arthropod-borne Pathogens ID IGEP Graduate Student Grant (PI) *2025*

Awards

- ✧ VT-CMB Travel Award *2025*
- ✧ Agee GTA Award for the best graduate teaching assistant *2024*
- ✧ Best Poster Award at VT Grado ISE Graduate Poster Competition *2024*
- ✧ ISE Grado Admission Travel Award *2022*

Honors

- ✧ Selected for the IISE Future Faculty Fellows (3F) Program *2025*
- ✧ Nominated and acknowledged as an outstanding undergraduate mentor at Virginia Tech *2024*
- ✧ Ranked in the top 10% of Industrial Engineering graduates, Sharif University *2021*
- ✧ Ranked in the top 1% in the Iran National University Entrance Exam *2016*

Software and Programming

- ✧ Languages (Python, C++); Solvers/Modeling (Gurobi, CPLEX, AMPL); Statistics (Minitab, SPSS); Simulation (AnyLogic); Other (Tableau, Visual Paradigm, SolidWorks)

Academic Services

- ✧ VT INFORMS President *2026 - 2027*
- ✧ VT ISC Graduate Student Grant Subcommittee Member *2026*
- ✧ VT INFORMS Vice President *2025 - 2026*
- ✧ Virginia Tech ISE Senior Symposium Poster Judge *2024 - 2025*
- ✧ Operations Research Liaison at Women in ISE (WISE), Virginia Tech *2024 - 2026*
- ✧ INCOSE Board Member at Virginia Tech *2023 - 2025*

Student Mentorship

✧ Kaiyu Yan, ISE PhD Student at Virginia Tech	<i>2025-Present</i>
✧ Seung-Jin Choi, ISE PhD Student at Virginia Tech	<i>2025-Present</i>
✧ Matthew Horne, ISE Undergraduate Student at Virginia Tech	<i>2026-Present</i>
✧ Matthew Asbury, ISE Undergraduate Student at Virginia Tech	<i>2026-Present</i>
✧ Joon Yong Shim, ISE Undergraduate Student at Virginia Tech	<i>2026-Present</i>
✧ Nihal Sageer, ISE Accelerated Master Student at Virginia Tech	<i>2024-2026</i>
✧ Joshua Cooper, ISE Undergraduate Student at Virginia Tech	<i>2025</i>

Professional Experience

✧ IT Manager, Pajooresh High School, Iran	<i>2019-2022</i>
✧ Project Manager, Karena Innovation Factory, Iran	<i>2020</i>